MG205: Econometrics Theory and Applications

Topic 3: Multivariate Regression Model

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The Real World Has Many Variables

All Models Are Wrong, Some Are Useful — George Box

We Cannot Capture Full Complexity

The Challenge We Face

- The world is **messy and interconnected**
- Everything affects everything
- We cannot *measure* all variables
- We cannot *include* all variables

How We Respond

- *Narrow down our model*
- Focus on specific relationships
- Make sensible, defensible assumptions
- Justify every variable choice

Models approximate reality to answer specific questions

Simple Regression Cannot Handle Confounders

Population Models: From One to Many Explanatory Variables

Simple Linear Regression

$$Y_i = \beta_0 + \beta_1 X_i + e_i$$

- **Clear interpretation:** β_1 is the slope
- Often too restrictive
- Suffers from omitted variable bias

Multiple Regression

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + u_i$$

- β_1 measures effect **holding X_2 constant**
- More realistic
- Addresses confounding

Exercise 1: What Does "Controlling" Mean?

Controlling Holds x_2 Fixed

Residualisation does not establish exogeneity

Holding constant

- How does the **fitted outcome** change with x_1, holding x_2 fixed?

Using remaining variation

- Fit x_1 using x_2 and an intercept.
- Use the residual part, uncorrelated with x_2 in this sample.

Separating the modelled term

- Including x_2 represents its linear contribution explicitly.
- This does **not** guarantee exogeneity or remove all confounding.

Exercise 2: Discuss the Interpretation of Multiple Regression

The Partial Derivative Holds x_2 Constant

Interpretation of the fitted regression

Fitted value

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2}.$$

Partial slope

$$\frac{\partial \hat{y}_i}{\partial x_{i1}} = \hat{\beta}_1.$$

- x_2 held fixed.
- Fitted coefficients held fixed.

Observed x_1 Is Its Linear Fit Plus a Residual

Define the auxiliary fit; centre it using MP3

Auxiliary OLS and its residual

$$\hat{x}_{i1} = \hat{a}_0 + \hat{a}_1 x_{i2},$$
$$\hat{v}_i = x_{i1} - \hat{x}_{i1}.$$

$$x_{i1} = \underbrace{\hat{x}_{i1}}_{\text{fitted part}} + \underbrace{\hat{v}_i}_{\text{residual part}}.$$

Centred decomposition: MP3

$$\hat{a}_0 = \bar{x}_1 - \hat{a}_1 \bar{x}_2,$$
$$x_{i1} - \bar{x}_1 = \hat{a}_1 (x_{i2} - \bar{x}_2) + \hat{v}_i.$$

No exact collinearity: $\sum_i \hat{v}_i^2 > 0$.

The x_1–Residual Cross-Product Is $\sum_i \hat{v}_i^2$

Auxiliary MP1 and MP2 remove the fitted part

Zero sample covariance

$$\sum_i \hat{v}_i = 0, \quad \bar{\hat{v}} = 0 \qquad \text{(MP1)},$$

$$\sum_i x_{i2}\hat{v}_i = 0 \qquad \text{(MP2)}.$$

$$\begin{aligned}
&\sum_i (x_{i2} - \bar{x}_2)(\hat{v}_i - \bar{\hat{v}}) \\
&= \sum_i x_{i2}\hat{v}_i - \bar{x}_2 \sum_i \hat{v}_i \\
&= 0.
\end{aligned}$$

Substitute the decomposition

$$\begin{aligned}
\sum_i \hat{v}_i x_{i1} =& \hat{a}_0 \sum_i \hat{v}_i \\
&+ \hat{a}_1 \sum_i x_{i2}\hat{v}_i \\
&+ \sum_i \hat{v}_i^2 .
\end{aligned}$$

$$\boxed{\sum_i \hat{v}_i x_{i1} = \sum_i \hat{v}_i^2} \qquad (1).$$

The Two Regression Residuals Are Orthogonal

Full-regression MP1 and MP2 justify each cancellation

Full-regression residual

$$\hat{\varepsilon}_i = y_i - \hat{y}_i,$$

$$\sum_i \hat{\varepsilon}_i = 0 \qquad \text{(MP1)},$$

$$\sum_i x_{i1}\hat{\varepsilon}_i = 0 \qquad \text{(MP2)},$$

$$\sum_i x_{i2}\hat{\varepsilon}_i = 0 \qquad \text{(MP2)}.$$

Substitute the auxiliary residual

$$\hat{v}_i = x_{i1} - \hat{a}_0 - \hat{a}_1 x_{i2}.$$

$$\begin{aligned}\sum_i \hat{v}_i\hat{\varepsilon}_i =& \underbrace{\sum_i x_{i1}\hat{\varepsilon}_i}_{0\ \text{(MP2)}} \\ & -\hat{a}_0 \underbrace{\sum_i \hat{\varepsilon}_i}_{0\ \text{(MP1)}} - \hat{a}_1 \underbrace{\sum_i x_{i2}\hat{\varepsilon}_i}_{0\ \text{(MP2)}} \\ =& 0 \qquad (2).\end{aligned}$$

Residualising Removes the Intercept and x_2 Terms

Multiply the fitted identity by $\hat{v}_i$; use (1) and (2)

Multiply and sum

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \hat{\varepsilon}_i.$$

$$\sum_i \hat{v}_i y_i = \underbrace{\hat{\beta}_0 \sum_i \hat{v}_i}_{0 \text{ (auxiliary MP1)}} + \hat{\beta}_1 \sum_i \hat{v}_i x_{i1} + \underbrace{\hat{\beta}_2 \sum_i \hat{v}_i x_{i2}}_{0 \text{ (auxiliary MP2)}} + \underbrace{\sum_i \hat{v}_i \hat{\varepsilon}_i}_{0 \text{ by (2)}}.$$

Substitute identity (1)

$$\sum_i \hat{v}_i x_{i1} = \sum_i \hat{v}_i^2.$$

$$\boxed{\sum_i \hat{v}_i y_i = \hat{\beta}_1 \sum_i \hat{v}_i^2} \quad (3).$$

The Simple-OLS Slope Equals $\hat{\beta}_1$

Regress the unchanged outcome on residual x_1 and an intercept

Second-stage OLS fit

$$y_i = \hat{c} + \hat{\theta}\hat{v}_i + \hat{r}_i.$$

Auxiliary MP1

$$\bar{\hat{v}} = 0, \qquad \sum_i \hat{v}_i = 0.$$

No exact collinearity: $\sum_i \hat{v}_i^2 > 0$.

Ordinary OLS slope

$$\begin{aligned}
\hat{\theta} &= \frac{\sum_i (\hat{v}_i - \bar{\hat{v}})(y_i - \bar{y})}{\sum_i (\hat{v}_i - \bar{\hat{v}})^2} \\
&= \frac{\sum_i \hat{v}_i y_i - \bar{y}\sum_i \hat{v}_i}{\sum_i \hat{v}_i^2} \\
&= \frac{\sum_i \hat{v}_i y_i}{\sum_i \hat{v}_i^2} = \hat{\beta}_1 \quad \text{by (3)}.
\end{aligned}$$

Exercise 3: CEO Compensation and Omitted Variable Bias

Firm Profits Affect CEO Salary, But What About Business Environment?

Understanding When Omitted Variables Create Bias

$$\log(\mathbf{salary}_i) = \beta_0 + \beta_1 \cdot \mathbf{profits}_i + e_i$$

- What about $\mathbf{environment}_i$ (quality of business environment)?
  - → The economy grows
  - → Business environment improves
  - → Firms make more profits
  - → CEOs have better outside options → must be paid more

Does omitting environment create bias? What is the direction?

Omitted Variable Bias Arises When Two Conditions Hold

Understanding the Bias Mechanism Through Assumptions

Consider the true model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + e_i$$

But we estimate

$$y_i = \gamma_0 + \gamma_1 x_{i1} + u_i$$

The omitted term enters the error

$$u_i = \beta_2 X_{i2} + e_i$$

For unbiasedness, we need

$$\mathbb{E}[u_i \mid X_{i1}] = 0$$

But this requires

$$\mathbb{E}[\beta_2 X_{i2} + e_i \mid X_{i1}] = 0$$

Which fails when

$$\mathbb{E}[X_{i2} \mid X_{i1}] \neq 0$$

This means X_{i1} and X_{i2} are correlated!

Visualising the Omitted Variable Problem

Dashed: auxiliary regression, not a causal effect



Omitted Variable Bias Requires Two Correlations

Both Conditions Exist in Our CEO Compensation Case

Omitted variable affects included variable

$$\text{cov}(\text{environment}_i, \text{profits}_i) > 0$$

- Good business environment → companies sell more
- Higher revenues → higher profits
- $\delta_1 > 0$

Omitted variable affects outcome

$$\text{cov}(\text{environment}_i, \log(\text{salary}_i)) > 0$$

- Better environment → CEOs have better outside offers
- Competition increases → firms pay more
- $\beta_2 > 0$

$\therefore$ Both correlations exist: omitted variable bias is present

Bias Is Positive, Our Estimate Is an Upper Bound

The Omitted Variable Bias Formula Reveals Direction

$$\text{Bias}(\hat{\beta}_1) = \underbrace{\beta_2}_{+} \times \underbrace{\delta_1}_{+}$$

- β_2: Effect of $\mathbf{environment}_i$ on $\log(\mathbf{salary}_i)$ in the full model
- δ_1: Effect of $\mathbf{profits}_i$ on $\mathbf{environment}_i$ in auxiliary regression

Economic Interpretation

$$\mathbb{E}[\hat{\beta}_1] = \beta_1 + \underbrace{(+)}_{\text{positive bias}}$$

- Our estimate $\hat{\beta}_1$ will be **too high**
- It's an **upper bound**, not a lower bound
- True effect of profits is **smaller** than our estimate

We cannot interpret $\hat{\beta}_1$ as a lower bound on β_1 because bias is positive

Exercise 4: Testing for Omitted Variable Bias Using Residuals

Sample Residuals Cannot Detect Bias

A Common Mistake in Applied Work

We ran simple OLS of y on x. Worried about omitted variable bias, we check:

$$\sum_{i=1}^{n} x_i \hat{\varepsilon}_i = 0$$

We find this holds in our sample.

Does this evidence suggest there is **no** omitted variable bias?
Answer: No, this evidence tells us nothing about omitted variable bias.

OLS Mechanics Do Not Establish Exogeneity

Unobserved errors and computed residuals are different objects

Concept	**Population**	**Sample**
Error / residual	e_i: unobserved model error	$\hat{\varepsilon}_i$: computed residual
Conditional mean	$\mathbb{E}[e_i \mid X_i] = 0$: assumption	Not established by residual orthogonality
Orthogonality	$\mathbb{E}[X_i e_i] = 0$: implied by exogeneity	$\sum_i x_i \hat{\varepsilon}_i = 0$: MP2
Across samples	Parameters fixed; errors can vary	Estimates and residuals can vary

Exercise 5: R^2 with Additional Variables

R^2 Never Decreases with More Variables

Why Model Fit Always Improves

$$R^2 = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = \frac{\text{Explained Sum of Squares}}{\text{Total Sum of Squares}}$$

With Fewer Variables

$$\text{earnings}_i = \beta_0 + \beta_1 \cdot \mathbb{1}[i \text{ is female}] + e_i$$

- All males get same $\hat{y}$
- All females get same $\hat{y}$
- Only 2 distinct predictions

With More Variables

$$\text{earnings}_i = \gamma_0 + \gamma_1 \cdot \mathbb{1}[i \text{ is female}] + \gamma_2 \cdot \text{education}_i + u_i$$

- Males differ by education
- Females differ by education
- Many more distinct predictions

Extra dimension allows better fit: $\hat{\varepsilon}_i$ smaller, R^2 higher (or equal)

More Variables Increase Coefficient Variance

The Cost of Controlling

$$\mathbf{var}(\hat{\beta}_j \mid \mathbf{x}) = \frac{\sigma^2}{n s_{x_j}^2 (1 - R_j^2)},$$

where $\mathbf{x} = (x_{i\ell})_{i=1,\ldots,n;\ \ell=1,\ldots,k}$ is the complete observed design, $s_{x_j}^2 = \sum_i (x_{ij} - \bar{x}_j)^2 / n$, and R_j^2 comes from regressing regressor j on the others. Index i labels observations; j labels coefficients.

When We Add x_2

- If x_1 and x_2 are correlated, then $R_1^2 > 0$
- $\therefore 1 - R_1^2 < 1$
- $\mathbf{var}(\hat{\beta}_1) \uparrow$

The Intuition

With correlated regressors:

- Harder to separate individual effects
- More uncertainty about each coefficient
- Need larger effects to reject H_0

This is the trade-off: better fit ($R^2 \uparrow$) but less precise estimates ($\mathbf{var}(\hat{\beta}) \uparrow$)

The Fundamental Trade-off: Fit vs Precision

Two Dimensions Cannot Be Simultaneously Optimised

Dimension	What It Measures	Impact of More Controls
Goodness of Fit	How well does model explain variation?	R^2 increases → better predictions
Hypothesis Testing	Can we detect effects?	$\text{var}(\hat{\beta}) \uparrow$ → wider confidence intervals → harder to reject H_0

The Dilemma

- **Adding variables**: Improves R^2 but increases $\text{var}(\hat{\beta})$
- **Dropping variables**: Risks omitted variable bias

Choose based on research question: Do you prioritise explanation or causal inference?

Exercise 6: Reducing Variance of Estimators

Only Sample Size Reduces Variance Reliably

What Can We Do About High Variance?

$$\mathbf{var}(\hat{\beta}_j \mid \mathbf{x}) = \frac{\sigma^2}{n s_{x_j}^2 (1 - R_j^2)}$$

Increase sample size ✓

- $n \uparrow$ directly
- $\Longrightarrow$ Variance $\downarrow$
- **Only reliable solution**

Choose x far from mean? ✗

- Would increase $s_{x_1}^2$
- Conflicts with random sampling (Assumption 2)
- Cannot control x values

Drop x_2? ✗

- Removes $(1 - R_1^2)$ term
- Creates bias if $R_1^2 > 0$
- Net effect ambiguous

Bias vs Variance Cannot Both Be Minimised

The Fundamental Tradeoff When Omitting Variables

Pros of Dropping Variables

- Lower $\mathbf{var}(\hat{\beta}_k)$
- Narrower confidence intervals
- Easier to detect effects

Cons of Dropping Variables

- Potential omitted variable bias
- Estimates may be misleading due to confounding
- Cannot interpret causally
- Worse model fit (lower R^2)

Think carefully about what matters most for your research question

Exercise 7: Low Variance and BLUE

Low Variance Concentrates Estimates

Why Estimator Variance Matters

Low Variance Estimator

- **Values across samples:** 0.08, 0.09, 0.10, 0.09, 0.08
- Concentrated around $\mathbb{E}[\hat{\beta}]$
- If also unbiased: close to β most of the time
- Reliable in practice

High Variance Estimator

- **Values across samples:** -0.50, 0.02, 0.80, -0.20, 0.15
- Widely dispersed, even if unbiased on average
- Might be far from β in any sample
- Unreliable in practice

Low variance + unbiasedness = estimates we can trust

OLS Has Lowest Variance Among Unbiased Estimators

The Gauss-Markov Theorem

Theorem 1 (Gauss-Markov Theorem) Under Assumptions 1-6, OLS is **B**est **L**inear **U**nbiased **E**stimator.

What Each Term Means

- *Linear:* Estimator is linear in y values
- *Unbiased:* $\mathbb{E}[\hat{\beta}] = \beta$
- *Best:* Lowest variance

Why This Matters

Among the **class of (all)** linear unbiased estimators, OLS has:

$$\mathbf{var}(\hat{\beta}_{\text{OLS}}) \leq \mathbf{var}(\hat{\beta}_{\text{other}})$$

OLS is optimal when assumptions hold - but **bias cannot be fixed with more data, variance can**

Putting It All Together

Model Selection Requires Theoretical Justification

Practical Guidance for Choosing Variables

Include Variables When:

1. **Theoretically important** for relationship
2. **Correlated** with both y and your x of interest
3. **Measurable** and available

Do NOT Include Just Because

1. They're available in data
2. They increase R^2
3. Someone else included them
4. They make results "look better"

Start with theory → specify model → justify choices → check robustness

What You Must Take Away from Today

The Five Key Lessons

1. **"Controlling"** has three equivalent interpretations: holding constant, using uncorrelated variation, removing from error term.
2. **Omitted variable bias** occurs when excluded variables correlate with included ones. Cannot be detected using sample residuals.
3. **Adding variables** reduces bias but increases variance. There is no free lunch.
4. **OLS is BLUE**: lowest variance among linear unbiased estimators under Assumptions 1-6.
5. **Two dimensions** (fit and precision) cannot both be optimised. Choose based on research goals.

Next Week

Next Class: Empirical Exercise 7

Daughters Change How Legislators Vote on Women's Issues

Paper: "Female Socialisation: How Daughters Affect Their Legislator Fathers' Voting on Women's Issues"
Author: Ebonya L. Washington
Journal: *American Economic Review*, 2008
Research question: Do daughters make fathers more feminist?
Method: Controlling for total children, each daughter increases legislator's propensity to vote liberally on reproductive rights.
Why it matters: Identifies child-to-parent behavioural influence in high-stakes decisions.
Reading week next week!